struggled, with only modest success, since the mid-1960s. Those questions, and our best 1993 answers (our "best guesses" is a better way to say it), are:

1. Does everything that enters the hole necessarily get swallowed by the singularity? We think so, but we're not sure.
2. Is there any route from inside the hole to another universe, or to another part of our own Universe? Very probably not, but we're not absolutely sure.
3. What is the fate of things that fall into the singularity? We think that things that fall in when the hole is quite young get torn apart by tidal gravity in a violent, chaotic way, before quantum gravity becomes important. However, things that fall into an old hole might survive unscathed until they come face-to-face with the laws of quantum gravity.

In the remainder of this chapter I shall explain these answers in more detail.

Recall that Oppenheimer and Snyder gave us a clear and unequivocal answer to our three questions: When the black hole is created by a highly idealized, spherical, imploding star, then (1) everything that enters the hole gets swallowed by the singularity; (2) nothing travels to another universe or another part of our Universe; (3) when nearing the singularity, everything experiences an infinitely growing radial stretch and transverse squeeze (Figure 13.1 above), and thereby gets destroyed. This answer was pedagogically useful; it helped motivate calculations that brought deeper understanding. However, the deeper understanding (due to Khalatnikov and Lifshitz) showed that the Oppenheimer–Snyder answer is irrelevant to the real Universe in which we live, because the random deformations that occur in all real stars will completely change the hole's interior. The Oppenheimer–Snyder interior is "unstable against small perturbations."

The Reissner-Nordstrom type of solution to the Einstein field equation also gave a clear and unequivocal answer: When the black hole is created by a particular, highly idealized, spherical, electrically charged star, then the imploding star and other things that fall into the hole can travel, via a "little closed universe," from the hole's interior to another large universe (Figure 13.4).